

Reproduction

1. Hormones of reproduction

2. Female reproduction:

- Functions
- Anatomy
- Oogenesis → *egg*
- Menstrual cycle → *prepare for pregnancy*
 - **Ovarian cycle:** changes in ovaries → egg
 - **Uterine cycle:** changes in endometrium → prepare for pregnancy
- Lactation

2. Male reproduction:

- Functions
- Anatomy
- Spermatogenesis → *sperm*

Hormones of Reproduction

hypothalamus-pituitary-gonadal axis

GnRH

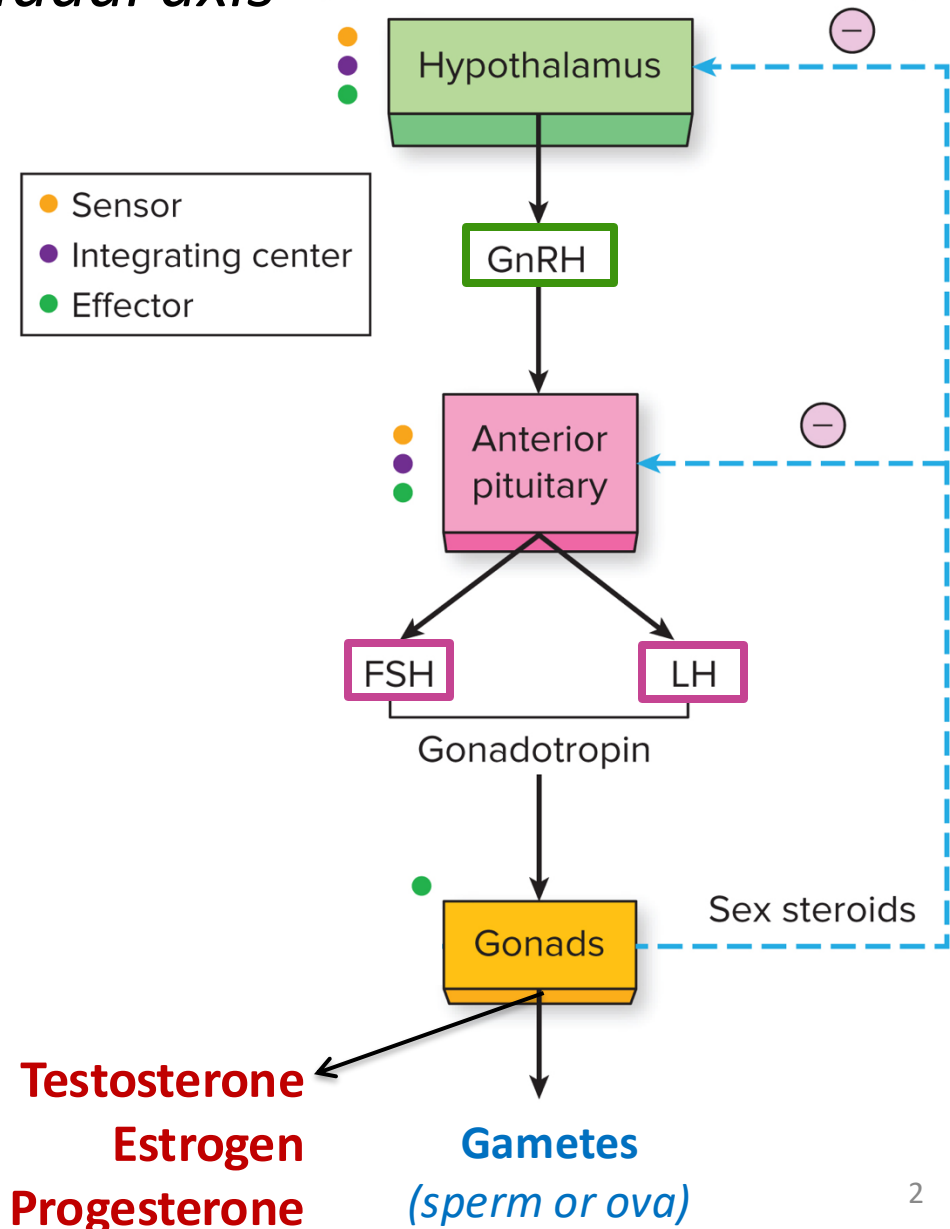
- *Gonadotropin-releasing hormone*

Gonadotropins

- **FSH**
Follicle stimulating hormone
- **LH**
Luteinizing hormone

Gonads:

- Exocrine: **gametes**
- Endocrine: **sex hormones (NFL)**



Female Reproduction

Functions:

1. Produce ova
2. Transport ova to site of fertilization & implantation
3. Preparation uterus for pregnancy
4. House & nourish the embryo/fetus
5. Move fetus to the outside
6. Produce female sex hormones

Anatomy Overview

Includes:

1. Gonads
2. Ducts
3. Accessory sex glands
4. External genitalia

Anatomy

gonad & ducts

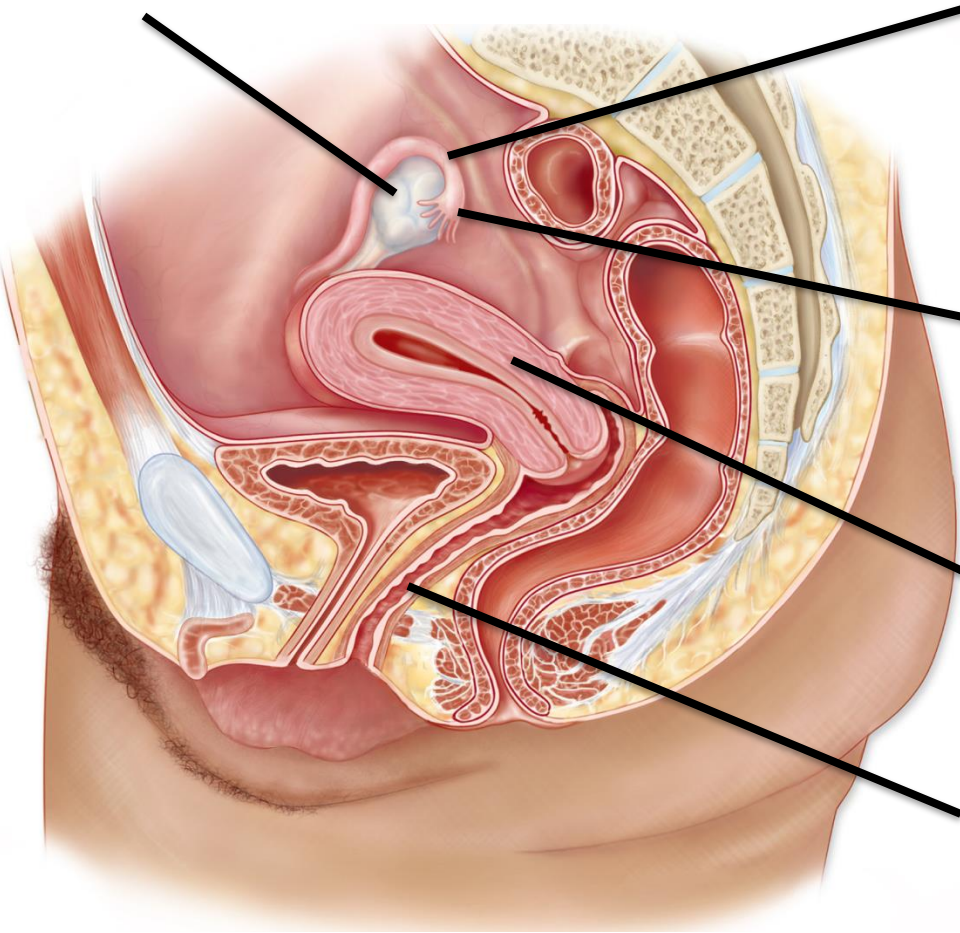
Ovary *produce gametes*

Fallopian tubes (*uterine tubes, oviduct*)
transport ovum; site of fertilization

Fimbriae (*beginning of duct*)
sweep in ovum (ciliated cells)

Uterus
house growing embryo/fetus

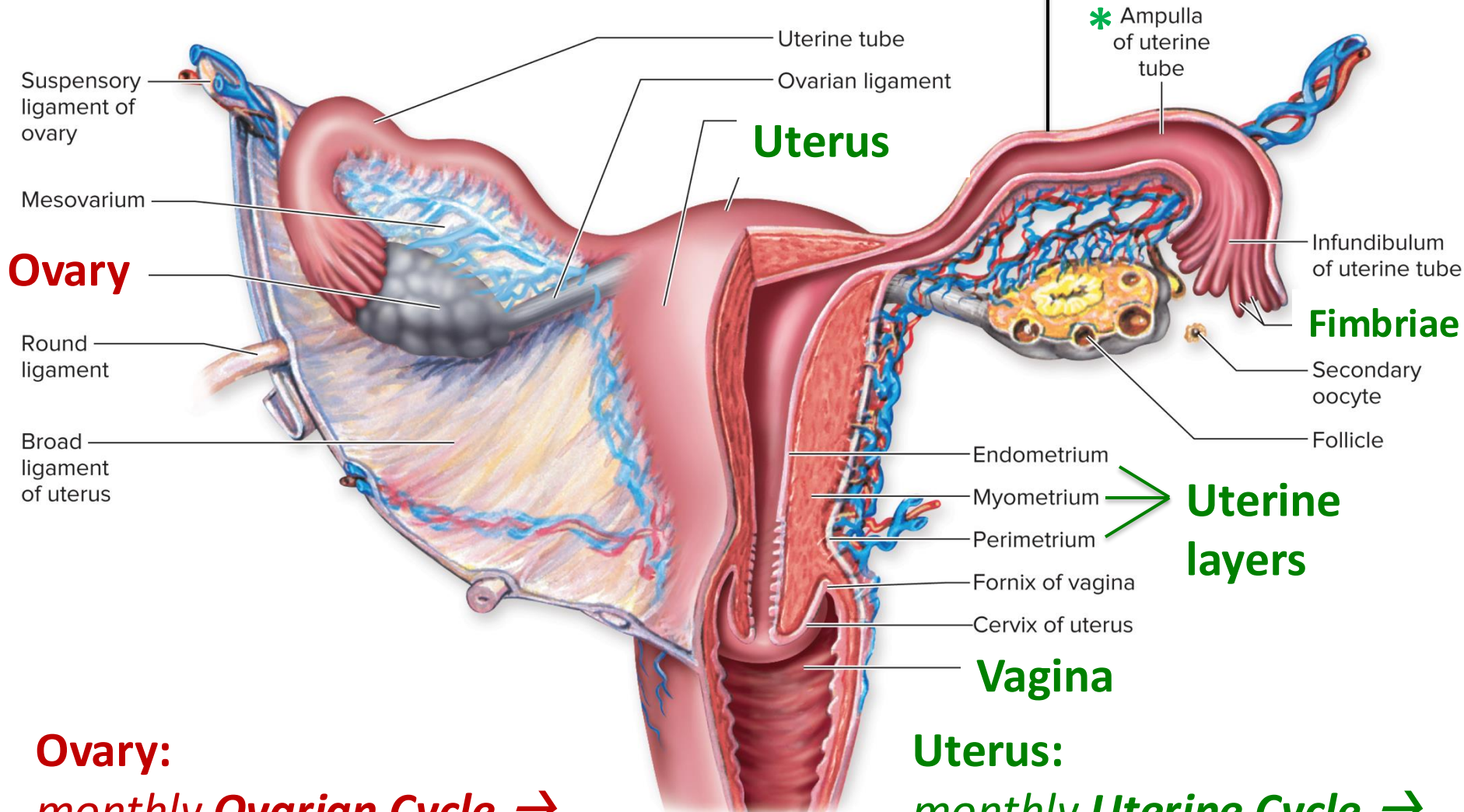
Vagina
receive sperm; deliver baby



Anatomy

gonad & ducts

Fallopian tubes



Ovary:

*monthly Ovarian Cycle →
1 ovum*

Uterus:

*monthly Uterine Cycle →
prepares for embryo*

Anatomy

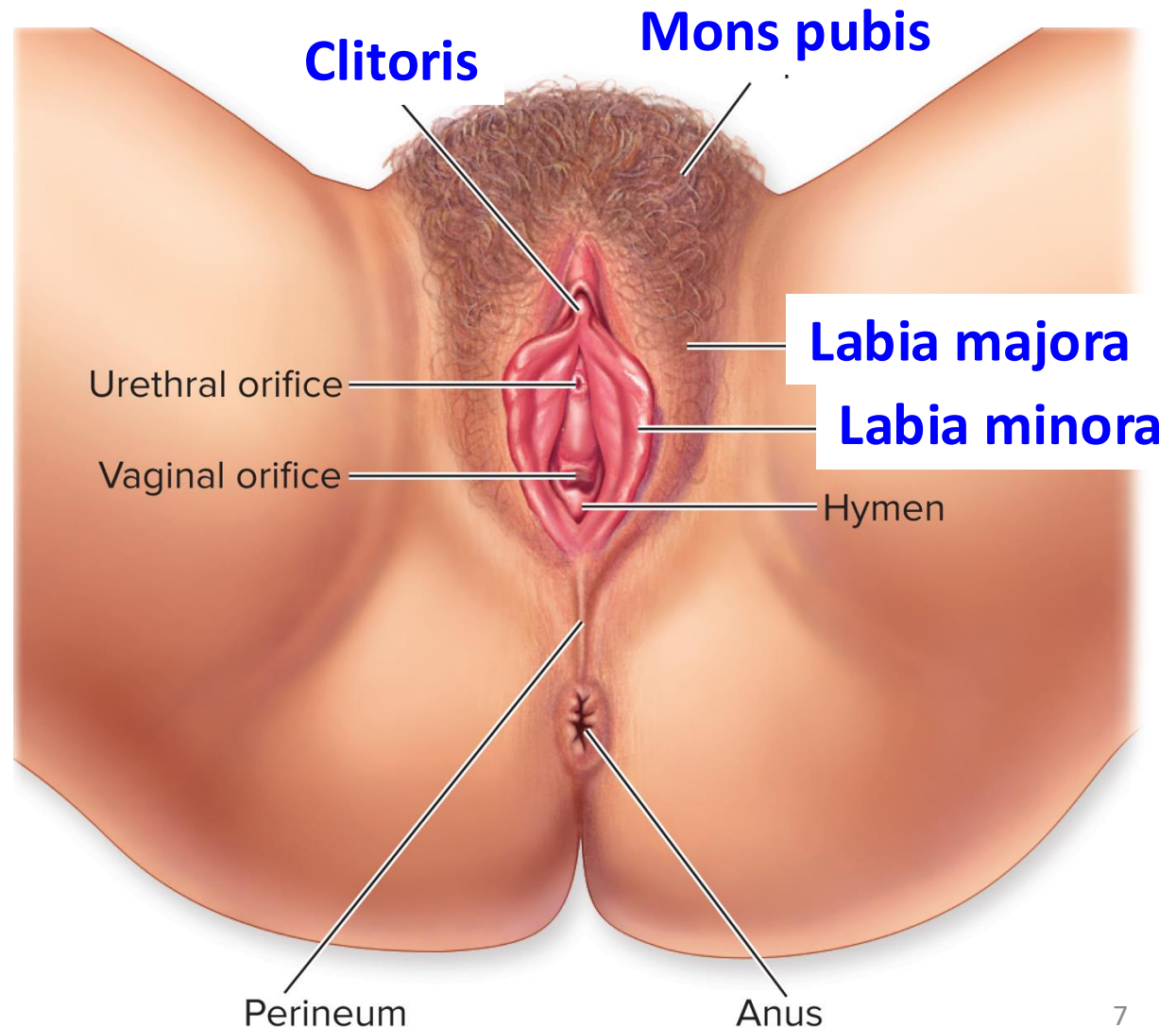
accessory sex glands & external genitalia/vulva

Greater vestibular/ Bartholin's glands:

*secretes lubricating fluid
outside of vaginal orifice*

External genitalia:

*covers/protects vaginal
and urethral orifices*



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Oogenesis

Cells are housed within
follicles (*not shown*)

Oogenesis

chromosomes

oogonium

46

Mitosis:
before birth

**primary
oocyte**

46

at puberty

primary oocyte

46

After puberty, one primary oocyte reaches maturity and is ovulated about once a month until menopause ensues

**secondary
spermatocyte**

(Just prior to ovulation)

**meiosis I
completes**

**arrested in
meiosis II**

**secondary
oocyte**

23

First
polar body

ovulation

(After fertilization)

**meiosis II
completes**

**mature
ovum**

23 x 2

Second
polar body

Polar bodies
disintegrate

Without fertilization
secondary oocyte is expelled

Zygote → mitosis
→ embryo

Meiosis

Hypothalamus-Pituitary-Gonadal Axis

GnRH

- Low in childhood
- Surge at puberty → sexual development
- Stimulates *anterior pituitary* → **FSH/LH**

FSH/LH

Initiates ovarian cycle:

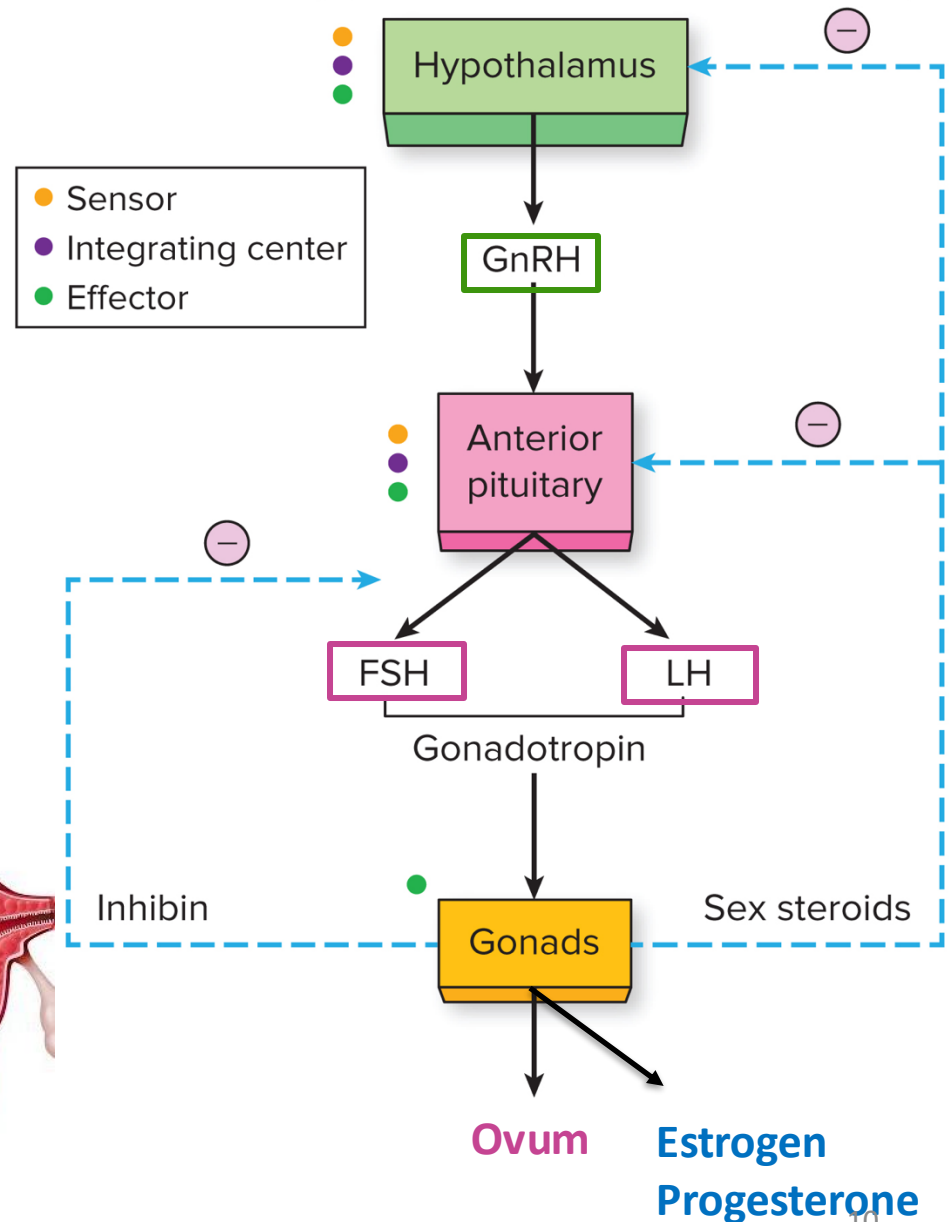
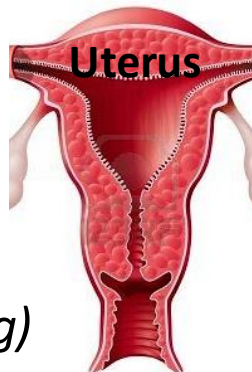
1. Oogenesis → ovum
2. Production of **estrogen & progesterone**



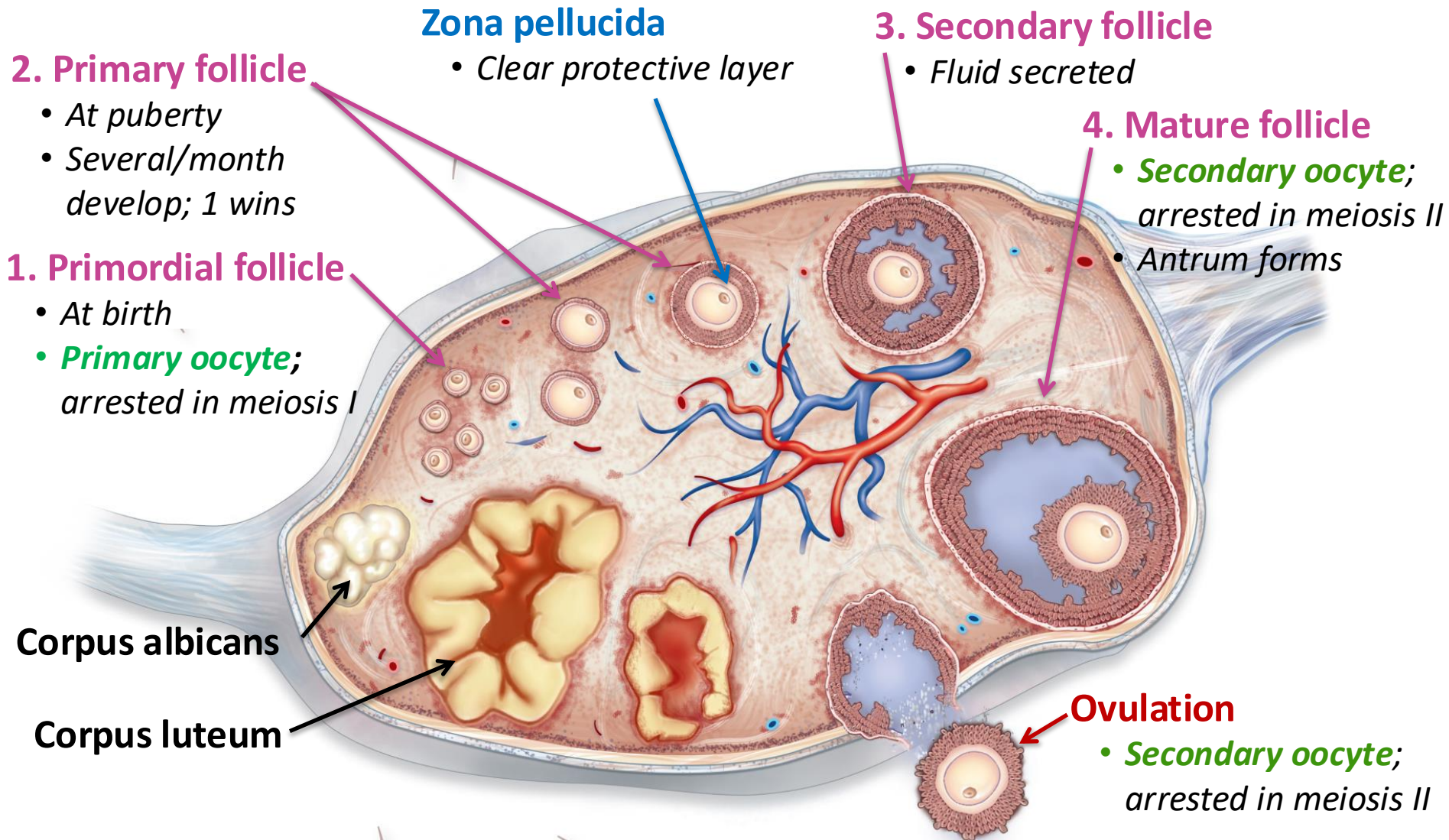
Estrogen/progesterone

Initiates uterine cycle:

1. Development of endometrium (uterus lining)



Ovarian Cycle: Follicle Development



As **follicle** develops:

- Surrounding cells divide + **zona pellucida forms** + antrum develops

Ovarian Cycle: Hormone Secretion

FSH stimulates follicle development.
(*follicular cells* become *granulosa cells* →
convert *testosterone* → *estrogen*)

LH → transforms connective tissue
→ *thecal cells* → *testosterone*

Granulosa cells secrete
nutrient rich *antral fluid*

Primordial follicle
follicular cells

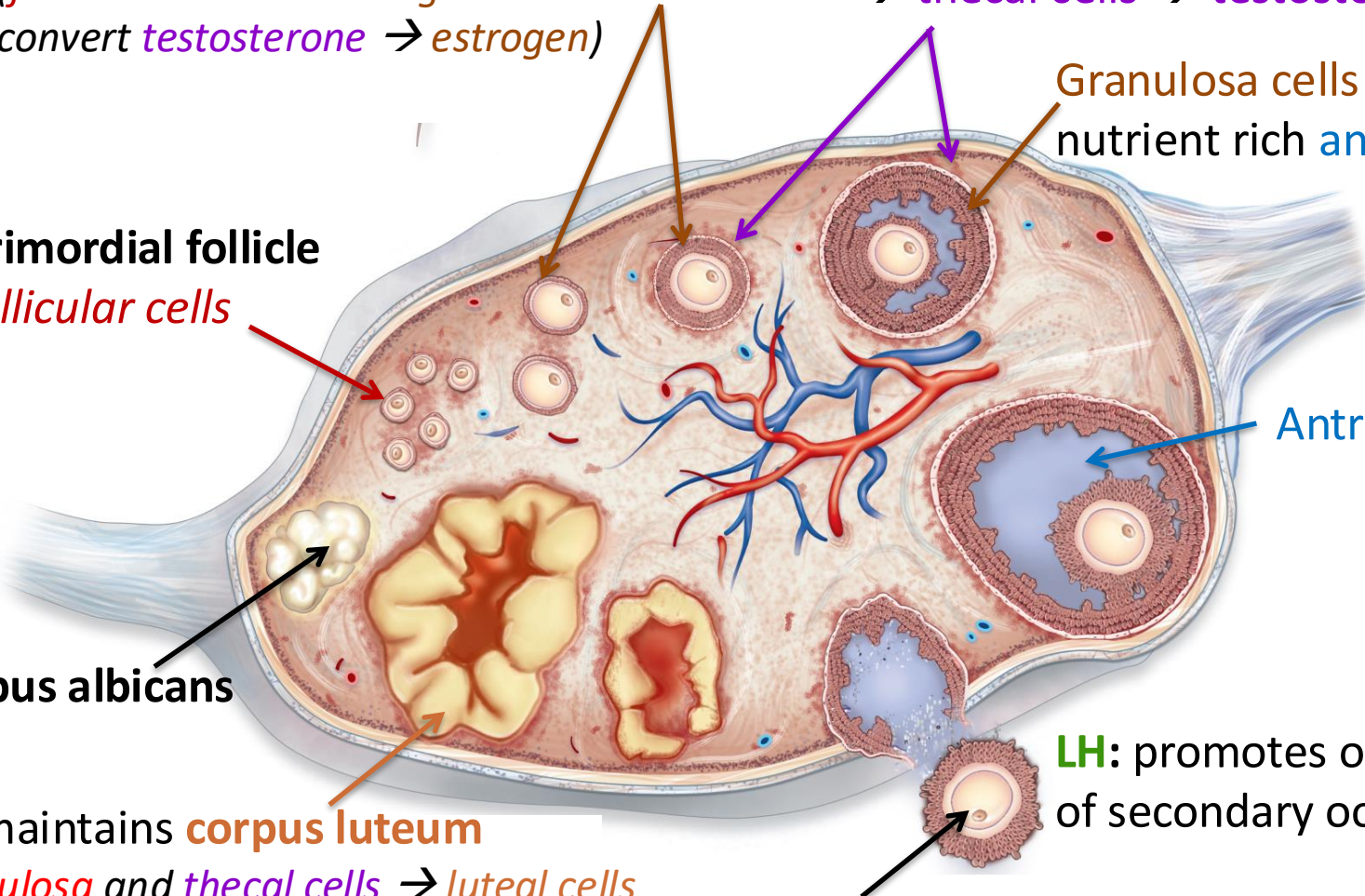
Antrum

Corpus albicans

LH: maintains **corpus luteum**
(*granulosa* and *thecal cells* → *luteal cells*
→ *estrogen + progesterone*)

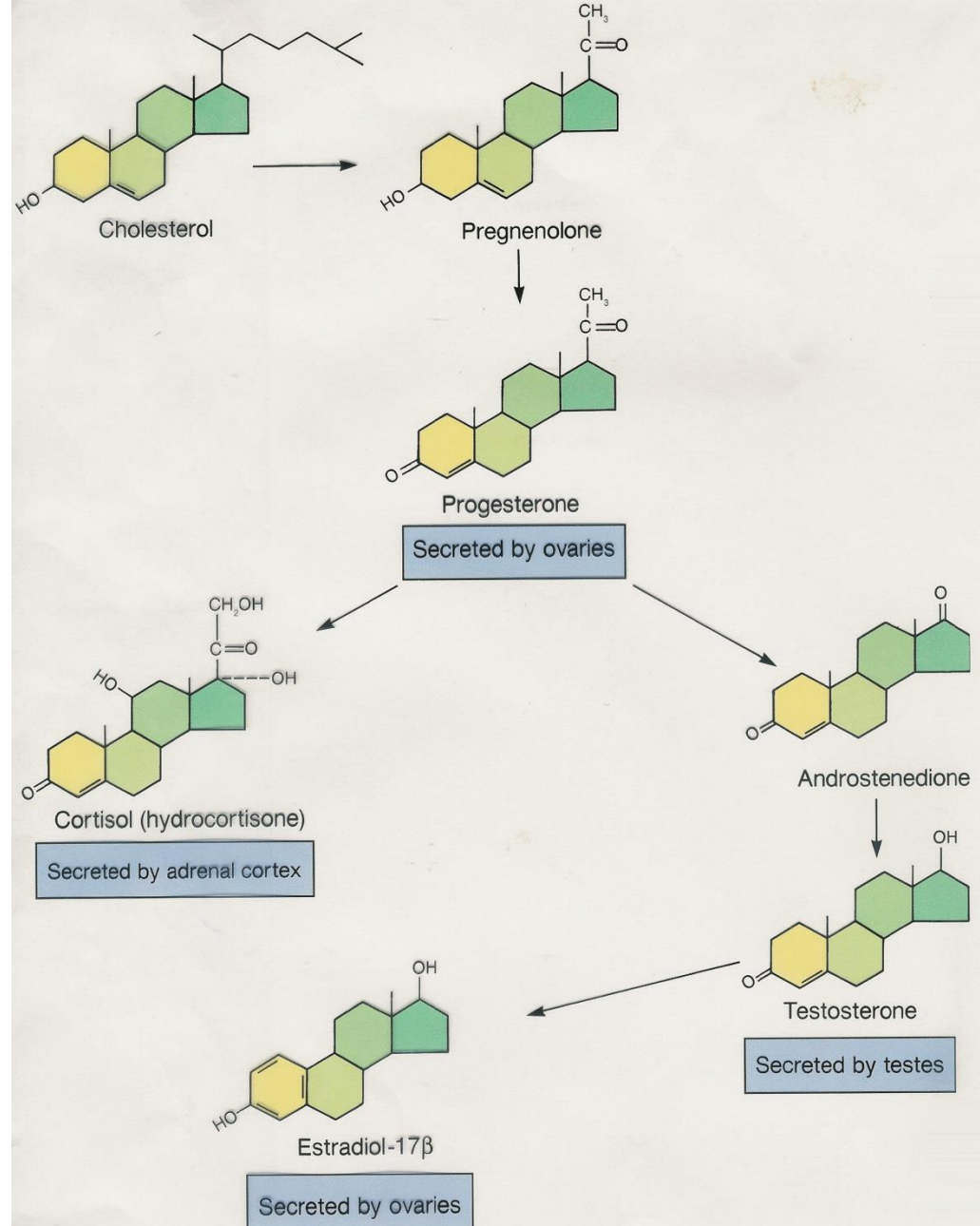
LH: promotes ovulation
of secondary oocyte

Secondary oocyte + sperm → **zygote**

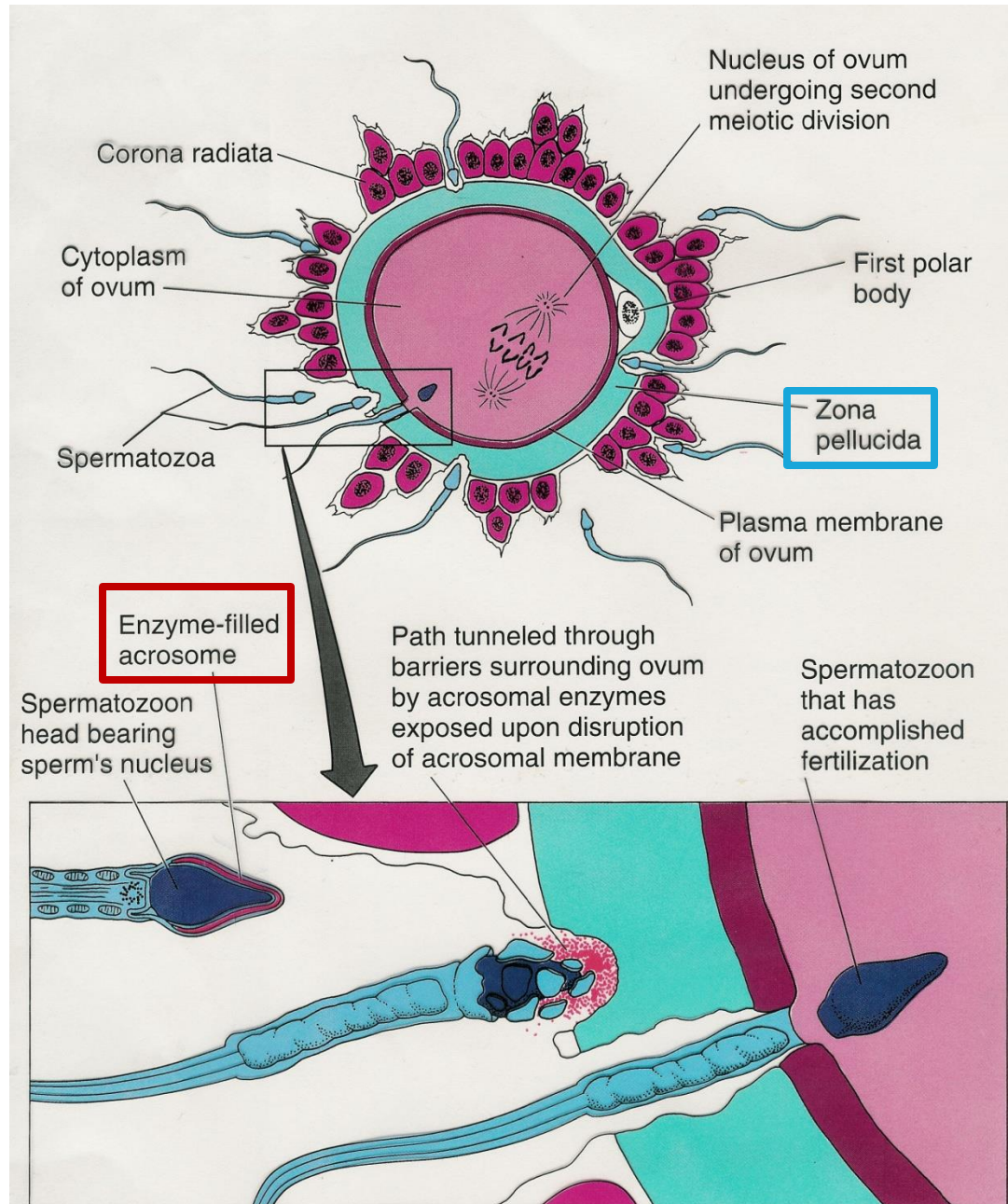


Synthesis of estrogen

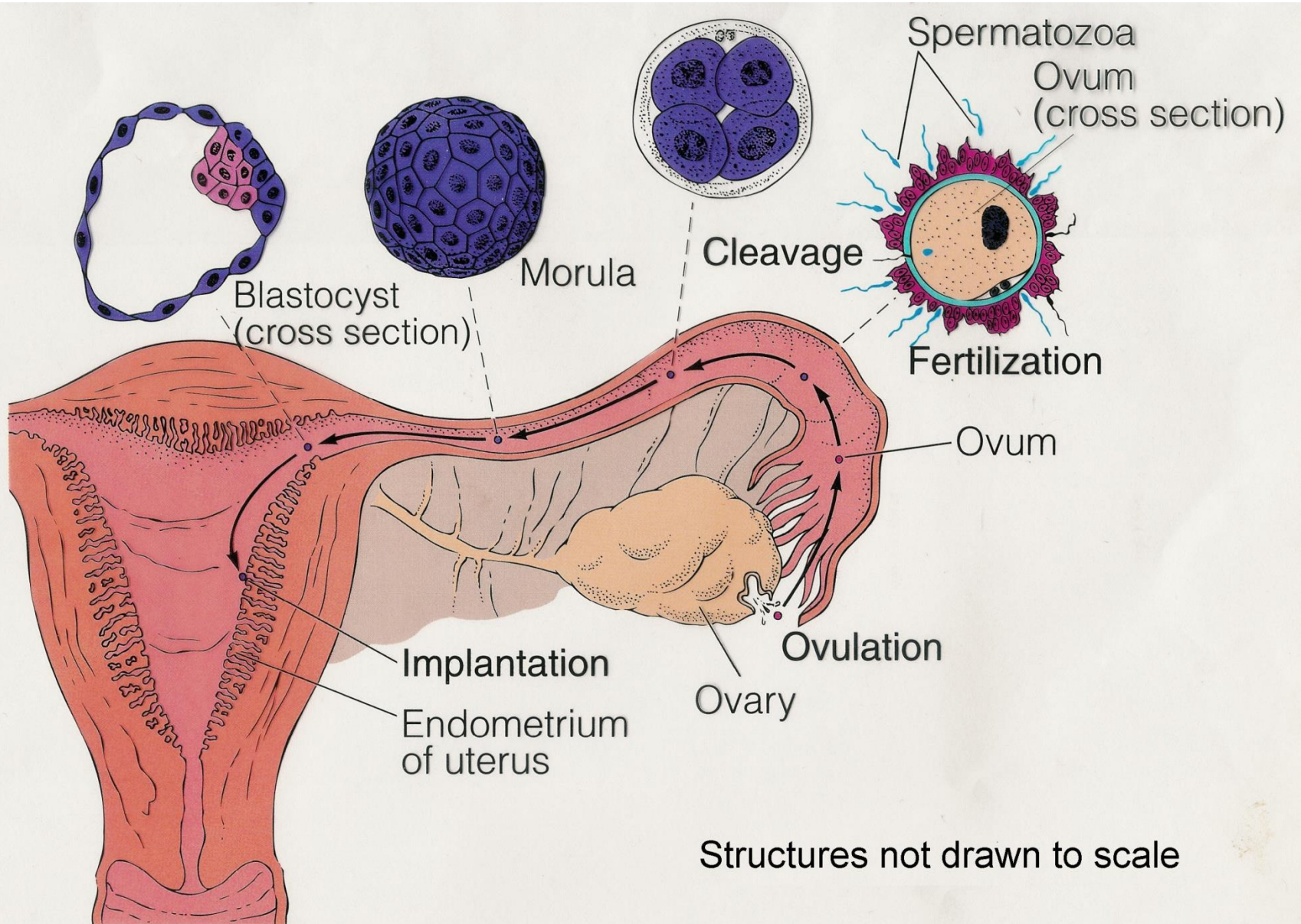
- **Thecal cells**
make testosterone
- **Granulosa/nurse cells**
convert testosterone to estrogen



Fertilization



Fertilization to Implantation



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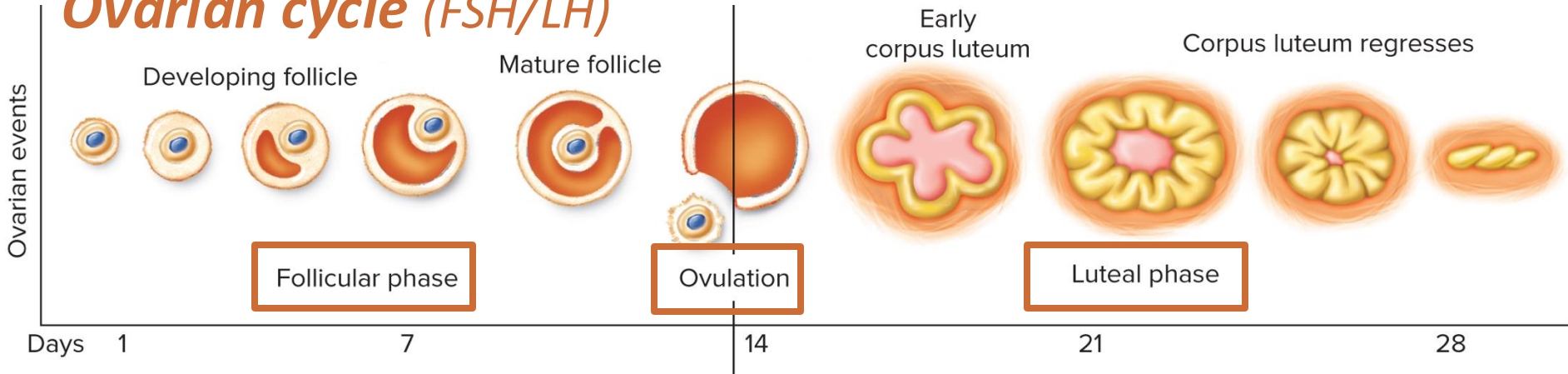
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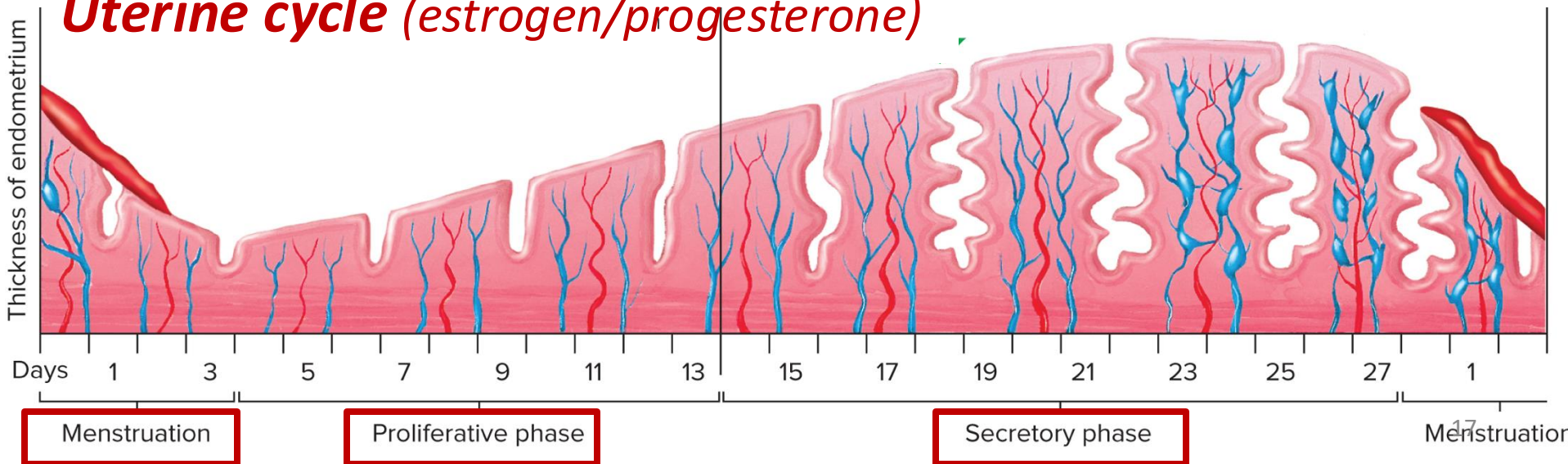
Phases

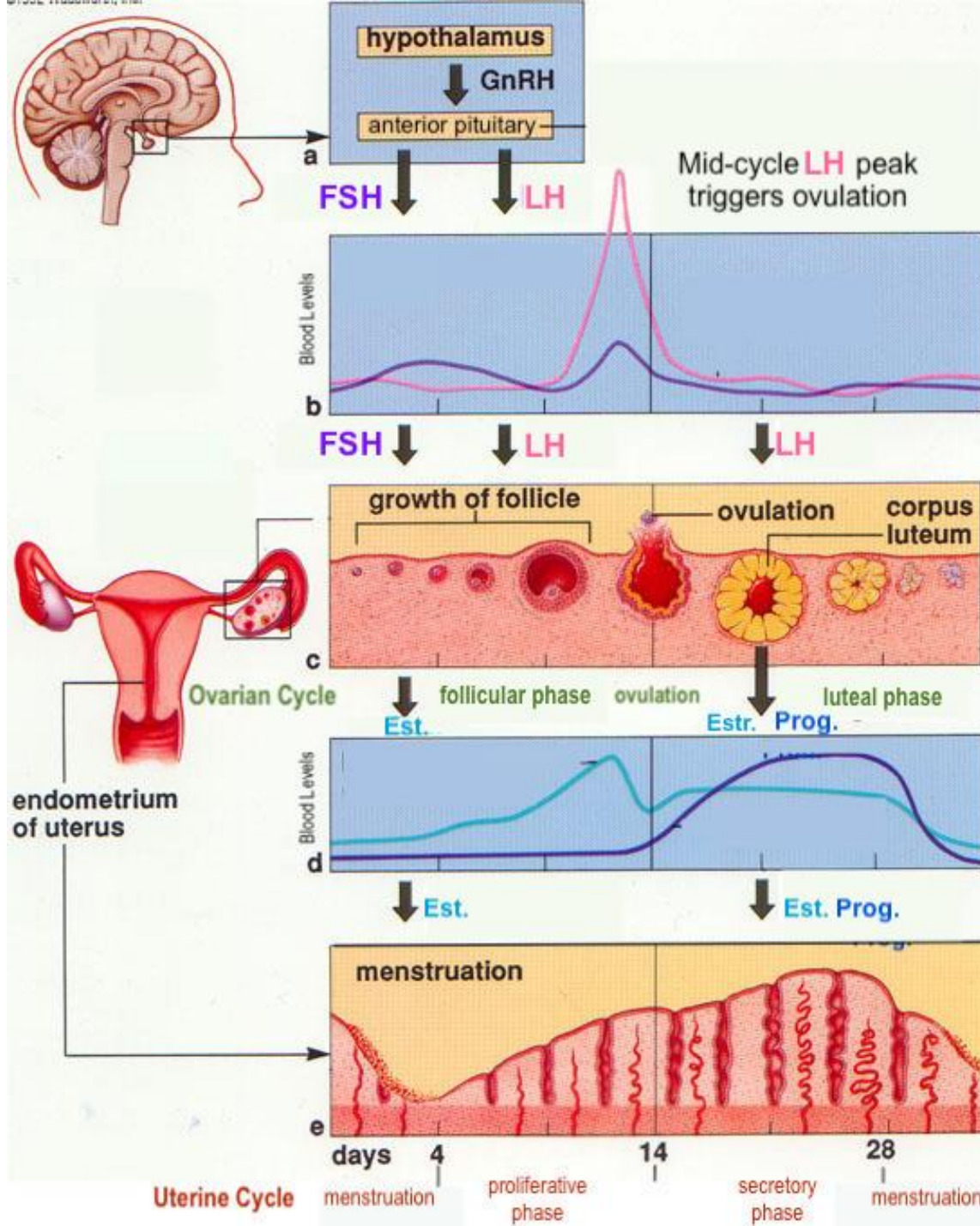
Menstrual cycle = **Ovarian cycle** + **Uterine cycle**

Ovarian cycle (FSH/LH)



Uterine cycle (estrogen/progesterone)

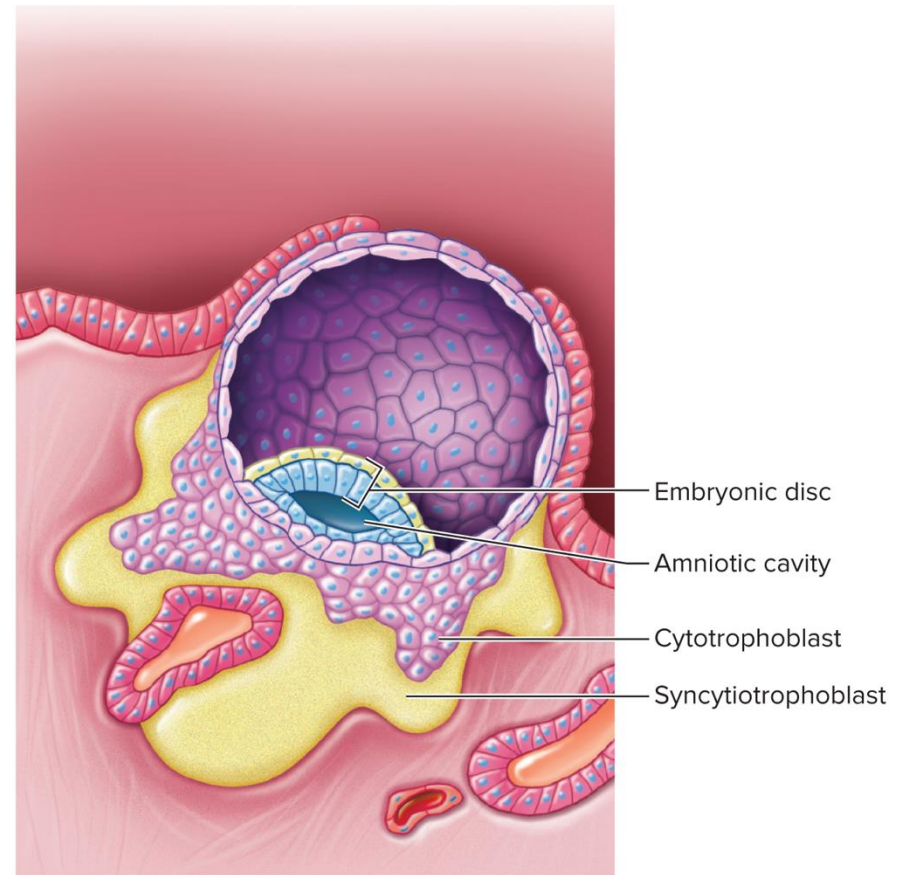
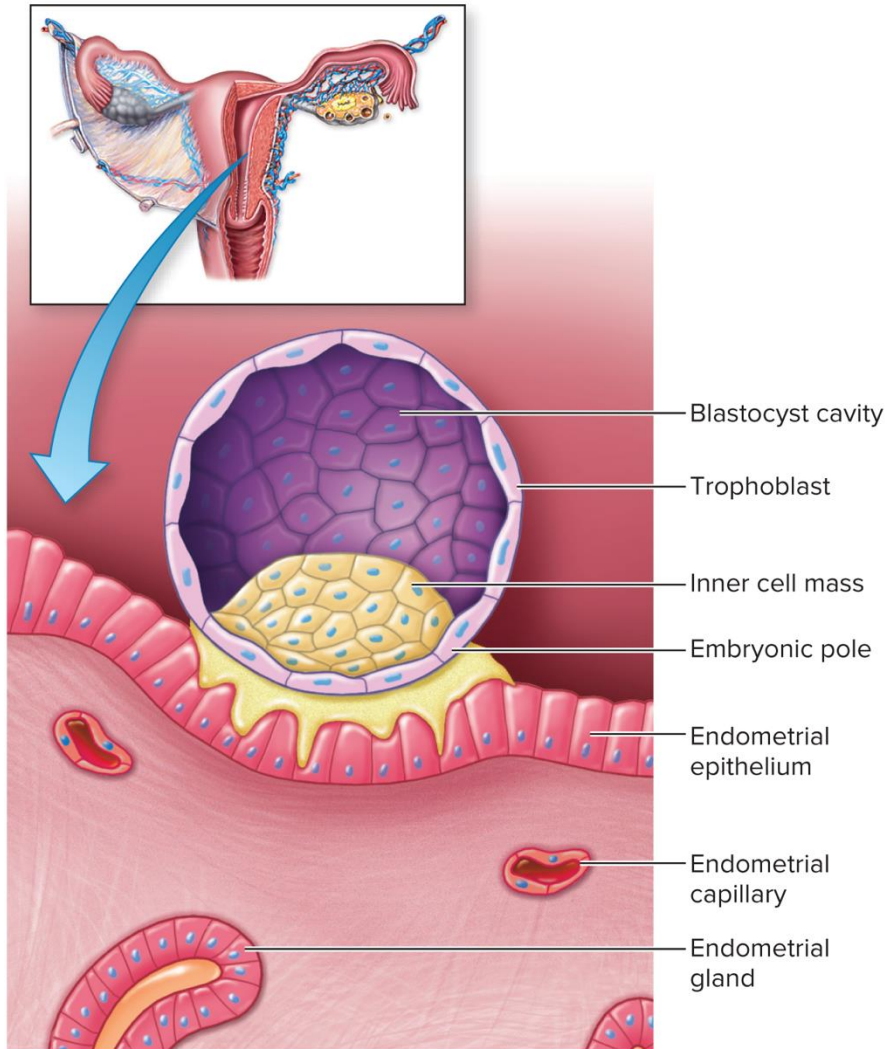




With pregnancy:

- Blastocyst cells makes **human chorionic gonadotropin (HCG)**
- **HCG** "saves" the corpus luteum (CL)
- Est. & prog. (CL) remain high supporting pregnancy
- Placenta eventually takes over Est. & Prog. production from CL
- CL regresses

Implantation of Blastocyst



(b)

Lactation: 3 Steps with Hormones

1. Build equipment

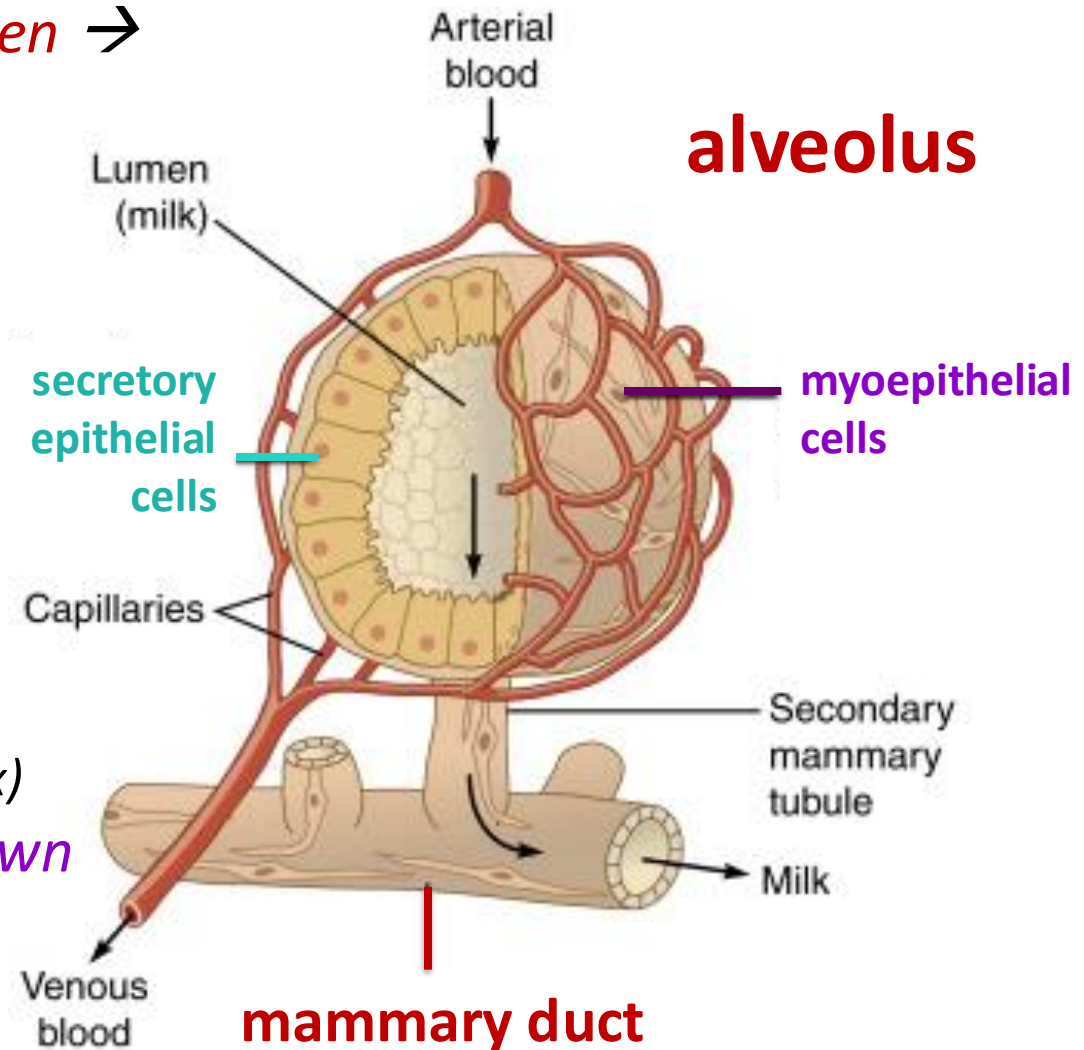
- *progesterone* & *estrogen* → *alveoli* & *ducts*

2. Make milk (blood filtrate)

- *prolactin* → *secretory epithelial cells*

3. Release milk (suckling reflex)

- *oxytocin* → *milk letdown*



Terminology to Know

- Menstruation – decline of E & P
- Ovulation – LH surge
- Fertilization
- Implantation – E & P prepare endometrium
- Gestation – E, P, hCG
- Parturition – oxytocin
- Lactation – E & P; prolactin; oxytocin